

4. Low Brightness Image

| *An image captured in low light appears very dark with poor visibility.*

(a) Interpretation of the scenario and cause of low brightness

An image captured under low-light conditions has most of its pixel intensities concentrated near the dark end of the grey-level scale (close to 0), producing a histogram skewed heavily toward low values. This happens because insufficient light reaches the camera sensor, often due to short exposure time, a small aperture, low ambient illumination, or sensor limitations in low light - resulting in an overall underexposed image with poor visibility of detail, especially in shadow regions.

(b) Key issues affecting image visibility

- Overall intensity is compressed into a narrow, dark range, hiding detail.
- Shadow regions lose almost all distinguishable detail (near-black clipping).
- Poor contrast between actual scene features, since they all sit in the same dark range.
- Visual quality is unsuitable for human viewing or for further processing like object detection.

(c) Applying an appropriate enhancement technique

A power-law (gamma) transformation is applied, using the formula $s = c \cdot r^y$ with $y < 1$ (e.g. $y = 0.4-0.6$). Because the exponent is less than 1, this transformation expands the darker intensity values over a wider output range while compressing the already-bright values less aggressively, effectively brightening the image and revealing shadow detail. Histogram equalization is a viable alternative that automatically redistributes the compressed dark intensities to use the full range.

(d) Justification

Gamma correction with $y < 1$ is preferred over a simple additive brightness shift ($s = r + k$) because a flat addition can clip bright pixels at 255 and does not proportionally recover the detail buried in dark regions. The non-linear power-law curve matches the way human vision perceives brightness (non-linearly), so it selectively boosts dark, detail-rich areas while leaving already-bright areas largely unaffected, avoiding over-exposure of highlights.

(e) Summary

Insufficient light during capture compresses pixel intensities into a dark, narrow range; the key issue is loss of shadow detail and poor overall visibility; and gamma correction ($y < 1$) addresses this effectively by non-linearly brightening dark regions while preserving highlight information, restoring a visually balanced image.